

AI Large Model Types and Principles

Course Content

This section introduces basic AI large model knowledge, as well as a brief overview of the types and principles of large models used in AI embodied intelligence applications.

AI Large Model Introduction

AI large models, also known as large-scale pre-trained models, are products of the deep integration of big data, massive computing power, and powerful algorithms. Simply put, they can be thought of as intelligent agents that have been "fed" with vast amounts of knowledge and repeatedly trained. By learning from large datasets, they master the patterns and rules within the data, thereby acquiring strong generality and generalization capabilities. This ability allows AI large models to no longer be confined to single tasks, but instead, like humans, flexibly apply their knowledge to solve complex problems across multiple domains. By learning data patterns from multi-source data such as internet text and images, they possess strong generalization capabilities, can adapt to various tasks through transfer learning or prompt engineering, and may even exhibit emergent abilities such as logical reasoning and commonsense understanding that were not explicitly programmed during pre-training.

Common AI Large Model Categories

Text Generation Large Models

- Based on the Transformer architecture, these models learn the grammar, semantics, and pragmatic rules of language through unsupervised or supervised learning on massive amounts of text data, enabling them to generate natural and fluent text based on input prompts or context. Pre-training through unsupervised learning makes them suitable for text generation and dialogue systems tasks; converting all text tasks into a unified text-to-text problem provides a more flexible framework that can handle translation, summarization, question answering, and various other tasks.
- In content creation, they can automatically generate practical texts such as articles, news, and comments, improving content production efficiency, and can also assist authors with creative brainstorming and text polishing.
- In the field of intelligent interaction, they can be applied to customer service and chatbots to generate natural and fluent responses, enhancing user experience.
- In personalized teaching, they can analyze problems, provide test point explanations, problem-solving approaches, and results, and can also help users with language learning.
- In machine translation, they can achieve automatic translation, and combined with speech models, can also achieve simultaneous interpretation, daily subtitle generation, and more.

Principle Introduction

Based on the Transformer architecture (especially self-attention mechanism), these models model the probability distribution and semantic associations of language through unsupervised/semi-supervised learning on massive text data, enabling natural language understanding and generation.

- Pre-training Logic

- Autoregressive (AR): Like the GPT series, predicting the next token through "causal language model" (e.g., "today's weather → today's weather is"), learning the forward-backward dependency relationships in text.
- Autoencoding (AE): Like BERT, predicting masked tokens through "masked language model" (e.g., "today [mask] is very sunny" → "weather"), learning bidirectional semantic context.
- Key Technologies
 - Attention Mechanism: Dynamically allocates weights to different words in text, capturing long-distance dependencies (e.g., "events mentioned earlier → impacts in later text").
 - Prompt Learning (Prompt Tuning): Activates specific capabilities of the model through templates (e.g., "Please summarize the following content: {text}"), adapting to downstream tasks.
- Emergent Abilities

As parameter scale expands (e.g., to hundreds of billions), models may exhibit emergent abilities such as logical reasoning, commonsense understanding, and few-shot learning that were not explicitly programmed during pre-training.

Multimodal Large Models

These models can process various types of input data, such as text, images, audio, video, and more. Through cross-modal learning, they understand the relationships between different modal data and integrate multimodal data together to fully utilize information from each modality, constructing a unified representation space where data from different modalities can mutually understand and combine with each other, thereby executing more complex and intelligent tasks. They can be used for cross-modal retrieval, retrieving data in one modality based on data in another; in visual question answering, models answer text questions based on image content; they can also generate image descriptions, producing natural language text describing image content; and enable multimodal dialogue involving multiple modalities of information, having broad application prospects in complex environments such as healthcare, transportation, and security monitoring.

Principle Introduction

Through cross-modal alignment and joint modeling, these models learn a unified representation space for different modal data, achieving semantic association and collaborative processing between modalities.

- Contrastive Learning: Like the CLIP model, mapping image and text feature vectors into the same space, training through "image-text pair matching" (e.g., "dog image → text 'dog'").
- Encoder-Decoder Architecture: Like DALL-E, text encoder extracts semantic features, image decoder generates corresponding images (text → image).

Fusion Methods

- Early Fusion: Merges multimodal data at the input layer (e.g., concatenating text embeddings with image pixel features).
- Late Fusion: Processes each modality data separately, fusing results at the decision layer (e.g., first independently analyzing text sentiment and image colors, then making a comprehensive judgment).

Speech Recognition Models

- Converting input speech signals into textual information. Usually based on deep learning algorithms, speech signals are first subjected to feature extraction, then features are input into neural network models for training and recognition. By learning from large amounts of speech data, the model recognizes different speech patterns and their corresponding textual content. They can assist customer service personnel in quickly recording customer needs and issues, improving service quality and facilitating subsequent queries; can be applied to voice search, freeing up hands, suitable for various search environments such as vehicle navigation and mobile phones; can also convert meeting conversations into text form, facilitating the organization and recording of meeting content; in human-computer interaction, they use voice commands to control intelligent facilities, including hardware facilities like robots and software applications.

Principle Introduction

Converting acoustic features of speech signals into text sequences, based on deep learning for end-to-end modeling.

- Feature Extraction: Preprocessing speech waveforms (such as framing, windowing), extracting Mel spectrograms (MFCC) or acoustic feature vectors (e.g., extracted via CNN).
- Sequence Modeling: Using Recurrent Neural Networks (RNN/LSTM) or Transformer encoders to capture temporal dependencies in speech sequences (e.g., "continuous syllables → words").
- Decoding Mapping: Mapping feature sequences to text sequences through Connectionist Temporal Classification (CTC) or attention mechanisms (e.g., acoustic features → "Hi,yahboom").

Speech Synthesis Models

- Converting input text into speech signals. Generally, models are trained to learn the mapping relationship between text and speech, generating corresponding speech features based on input text, then converting features into audible speech through speech synthesis technology. Widely used in areas such as voice assistants, audiobooks, and intelligent customer service, providing speech interaction services to users and enabling devices to communicate with users in natural and fluent speech.

Principle Introduction

Converting text semantics into natural and fluent speech signals, simulating the rhythm, intonation, and emotion of human speech.

Deep Learning Synthesis:

- Text Analysis: Parsing text semantics, part of speech, and emotion through NLP models (e.g., "I'm very happy today" → cheerful tone).
- Acoustic Modeling: Using Tacotron series (encoder-decoder + attention mechanism) to generate Mel spectrograms of speech.
- Vocoder: Converting Mel spectrograms into waveform signals, such as WaveNet and HiFi-GAN (improving speech naturalness).

AI Model Comparison Summary

Model Type	Input	Output	Core Technology	Typical Scenarios
Natural Language LM	Text	Text	Transformer Self-Attention	Writing, Dialogue, Translation
Multimodal LM	Text + Image/Audio	Cross-modal Content	Cross-modal Alignment, Joint Encoding	Image-text Generation, VQA
Speech Recognition	Speech Waveform	Text	Acoustic Feature Extraction + Seq Decoding	Meeting Transcription, Voice Search
Speech Synthesis	Text	Speech Audio	Text Analysis + Acoustic Modeling + Vocoder	Voice Assistant, Audio Content Production